

CMUT Array for Deep Vein Thrombosis

Sören Köble¹, Nicolas Lange¹, Marco Kircher¹, Uwe Völz¹, Sandro G. Koch¹

¹Fraunhofer-Institut für Photonische Mikrosysteme, Dresden, Germany

Background, Motivation and Objective

Recent advances in medical diagnostics increasingly focus on online health monitoring. Established use cases include continuous recording of vital signs such as heart rate, blood pressure, and electrocardiograms. However, to enable early detection of critical health conditions like deep vein thrombosis (DVT), additional information on blood flow is required.

Ultrasound-based flow measurement is a common method for this type of diagnosis, but it is currently limited to clinical environments. For high-risk patients, such as those with thrombosis or cardiac insufficiency, there is a strong need for solutions that are easy to use in everyday life and do not impair quality of life. This calls for compact, wearable devices with integrated sensors, electronics, and intelligent AI-based signal processing and evaluation.

To enable integration into such medical wearables, our work focuses on reducing the required CMUT drive voltage while maintaining operating frequency and signal-to-noise ratio.

Statement of Contribution/Methods

Capacitive Micromachined Ultrasonic Transducers (CMUTs), characterized by high receive sensitivity and broad bandwidth, are well suited for classical sonography, particularly because of their inherent acoustic matching to water. At Fraunhofer IPMS, advances in CMUT structure optimization have enabled the realization of small apertures and large arrays with operating frequencies up to 30 MHz. CMUTs produced on silicon wafers allow direct integration of analogue and digital CMOS circuits, enabling highly integrated ultrasonic front ends.

The required drive voltage of CMUTs scales predominantly with the electrode gap. We therefore targeted a reduction of the gap height to lower the drive voltage. This required overcoming several technological challenges, achieving a highly uniform and precisely controlled cavity height during the polishing process.

Because the vibrating plate cannot move more than the cavity height, reducing the gap also limits the maximum transmitted acoustic pressure. Simultaneously this increases electro-mechanical coupling and thus higher sensitivity of the CMUT at smaller gaps, resulting in similar SNR.

Results/Discussion

We present a CMUT design optimized for wearable applications by reducing the electrode gap. The required drive voltage was decreased from more than 120 V to 45 V while preserving the target operating frequency and signal-to-noise ratio. This reduction is particularly advantageous for medical wearables, enabling the integration of CMUT-based ultrasound sensing into compact, battery-powered devices for continuous patient monitoring and ambulatory care.